Compositional Verification of Abstract Interpretations Using Multi-Language Semantics

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which calls can apply which λ?

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$$\{(\ell_1, c_2), (\ell_3, c_3), (\ell_2, c_1)\}$$

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  The set of violated contracts / blames.

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Trace Events

Collecting Semantics

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- Instrument a standard concrete semantics

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- Most precise

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- Instrument a standard concrete semantics
- Most precise
- Uncomputable

Collecting Semantics: Uncomputable

$$\varsigma_1 \rightarrow \varsigma_2 \rightarrow \varsigma_3 \rightarrow \varsigma_4 \rightarrow \varsigma_5 \rightarrow \varsigma_6 \rightarrow \varsigma_7 \rightarrow \varsigma_8 \rightarrow \varsigma_9 \rightarrow \cdots$$

Collecting Semantics: Uncomputable

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Abstract to a *finite* representation

$$\begin{array}{ccccccc} \widetilde{\varsigma_a} & \longrightarrow & \widetilde{\varsigma_b} & \longrightarrow & \widetilde{\varsigma_e} & \longrightarrow & \cdots \\ & \swarrow & \uparrow & & & & \\ \widetilde{\varsigma_c} & \longrightarrow & \widetilde{\varsigma_d} & & & & \end{array}$$

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Abstract Semantics

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- Abstract Abstract Machines (Van Horn and Might 2010)

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All concrete program traces are covered by some abstract trace.

Separate Compilation

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let add1 = fun n => n + 1 in
let g = fun m => 2 * (add1 m) in
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(h 3)
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What can we say about `M.f` and `h`?

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Does • really mean **any value**?

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- Link `M.f` with `id`

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- λ bound by g has escaped!

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(* Module M *)
let f = fun _ =>
 let sq = fun x => x * x in
 sq
```

- Link `M.f` with some constant function

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(* f is defined in some module M *)
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What about the λ bound by add1?

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- Can `(M.f g)` ever return it?

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- `(M.f g)` should never return it!
- It should never be called when we apply `h` to 3
- • does **not** include internal non-escaping values

- are exactly the external values + the internal escaping values.

- Imports are •

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- $[c]$ $(\bullet$ $\mathtt{v})$

- Imports are •
- [c] (• v)

  - (• v) -> •

- Imports are $\bullet$
- $[c]$ `(•  v)`
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Please note that $\bullet$ is opaque

Shivers' Escape Technique 1991

- Much finer-grained
- Work with the set of escaped values instead of $\bullet$

Midtgaard, Adams, and Might 2012

- **Internal** vs **External**

Midtgaard, Adams, and Might 2012

- **Internal** vs **External**
- Galois connections

Midtgaard, Adams, and Might 2012

- Mechanizations and verified compilation

Midtgaard, Adams, and Might 2012

- Mechanizations and verified compilation
- Generalization to other trace events

Midtgaard, Adams, and Might 2012

- Logical relations based soundness proofs

Midtgaard, Adams, and Might 2012

- Logical relations based soundness proofs
- Need a generic way to differentiate the **trace events**

Multi-Language Semantics

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- Gradual typing

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- Gradual typing
- Foreign function interfaces

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$$(\ell_1, c_2), (\ell_3, c_3), (\ell_2, c_1)$$

```
(* Main *)
let add1 = ℓ1fun n => n + 1 in
let g = ℓ2fun m => 2 * c1(add1 m) in
let h = c2(M.f g) in
c3(h 3)
```

```
(* Module M *)
let f = ℓ1fun g' =>
 let sq = ℓ2fun x => x * x in
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$$\{ (\ell_\bullet, c_2), (\ell_\bullet, c_3), \qquad , (\ell_2, c_\bullet), \qquad \}$$

- Boundary

$$\{\ (\ell_1, c_2), (\ell_3, c_3), (\ell_2, c_1), (\ell_2, c_2), (\ell_1, c_1)\ \}$$

$$\{\ (\ell_\bullet, c_2), (\ell_\bullet, c_3), (\ell_\bullet, c_\bullet), (\ell_2, c_\bullet), \qquad\qquad \}$$

- Boundary
- External

$$\{ (\ell_1, c_2), (\ell_3, c_3), (\ell_2, c_1), (\ell_2, c_2), (\ell_1, c_1) \}$$

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- Boundary
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Proof Sketch

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